

ME1D01: Engineering and Environmental Management
with English Writing Requirements

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Waste Reduction Policies in Hong Kong

Word count: 2458 words

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Date: **03/12/2023**

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Introduction

Hong Kong (HK) is one of the most waste-producing cities in the world. The most recent waste report by the Hong Kong Environmental Protection Department (EPD) in 2021 states that the total quantity of solid waste disposed of at strategic landfills was at a record high of 5.67 million tonnes. The average daily amount was 15,533 tonnes per day. [1] In addition, HK's waste is disposed of at three landfills. This phenomenon is highly concerning in an environmental context, as the landfill pollutes its surroundings due to climate change and health hazards. As such, the HK Government has implemented historical measures to reduce waste to relieve the stress of the high amount of trash that the city places on the landfills. *Waste reduction* is the minimisation of waste from the source before processing. Different methods can facilitate this, as seen by the waste hierarchy pyramid below:

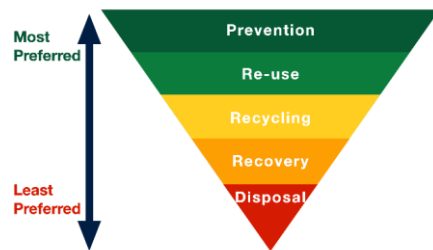


Figure 1: Diagram of Waste Hierarchy [2]

The Waste Hierarchy indicates the order of preference that humanity should take to manage their waste, sorted from top to bottom. In the context of HK, most of the waste is at the “Disposal” level, the lowest level of the Waste Hierarchy, which intensifies the need for solutions to reduce waste.

Observing HK's increasing municipal solid waste (MSW) output throughout the past few decades, it is irrefutable that HK requires practical solutions to reduce its waste. HK has implemented numerous waste reduction policies to reduce trash disposal in the past decade. As such, this report aims to examine the efficacy of past approaches to analyse how they lower the daily garbage output in correlation with the increasing solid waste disposed to identify the

reasons for the increase in solid waste year after year. Specifically, this report will focus on the schemes based on HK's "Polluter-Pays" principle.

This report is separated into four sections: discussing problems HK is currently facing relating to waste, examining current policies that HK is utilising, inspecting the future policies that HK will be implementing, and finally, giving recommendations to improve the procedures discussed in this report.

Summary of Hong Kong's Waste Management

Municipal Solid Waste Disposal

According to the EPD, 69% of MSW is disposed of in landfills in 2021. The remaining 39% is recovered through recycling or treatment. Despite this, HK's landfills have been taking MSW at a high number. Li et al. [3] state that landfill problems have been a concern since the 2010s, and the landfills are nearing saturation with the vast amount of waste. The central reason that landfills are nearing saturation can be attributed to the large amount of garbage that the HK population produces. Wong [4] states that HK has the highest amount of domestic waste per capita compared to London and nearby cities, including Tokyo, Taipei City, and Seoul, at a similar level of development.

Waste Management

With the alarmingly high amount of waste output that HK produces daily and the gradual saturation of HK's landfills, HK needs to reduce its waste. To achieve this goal, the HK government has drafted the "Waste Blueprint for Hong Kong 2035", highlighting two main medium-term objectives of the EPD: reducing the per capita MSW disposal by 40-45% and increasing the waste recovery rate to about 55% [5], which extends to the Hong Kong Climate Action Plan's long-term targets of zero municipal waste by 2035 and carbon neutrality by 2050 [6].

Hong Kong currently has three implemented waste reduction policies, and two will be implemented to reduce the waste produced by the HK population. These policies all follow the principle of “polluter pays”, a waste-charging scheme that encourages polluters to reduce their waste emissions. The Organisation for Economic Co-operation and Development (OECD) defines “polluter pays” as a principle where polluters should bear the expenses of carrying out the pollution prevention and control measures introduced by public authorities to ensure that the environment is in an acceptable state [7].

The three “polluter pays” policies currently implemented are Producer Responsibility Schemes (PRS), which are listed below:

1. Plastic Shopping Bag (PSB) Charging Scheme
2. PRS on Waste Electrical and Electronic Equipment (WEEE)
3. PRS on Glass Beverage Containers

The two that will be implemented are listed below:

1. Municipal Solid Waste (MSW) Charging Scheme
2. PRS on Plastic Beverage Containers

This report will focus on one presently enforced PRS and another implemented in the future: the PSB Charging Scheme and the MSW Charging Scheme.

PSB Charging Scheme

Overview of the PSB Charging Scheme

The EPD ran a pilot for the PSB Charging Scheme in 2009, targeting 3,000 retail outlets, primarily supermarkets, convenience stores and medicine and cosmetics stores [8]. Full implementation was done in 2015, whereby plastic bags are charged \$0.5, excluding some outlying cases. On December 31, 2022, the third and latest update for the PSB Charging Scheme increased the charging level to 1 HKD and tightened regulations regarding exemptions.



Figure 3: Sticker for Enhanced Scheme for PSB Charging [9]

Effectiveness and Public Reception

The effectiveness of the PSB Charging Scheme can be seen after its full implementation in 2015; the total number of PSBs disposed of at landfills in 2015 dropped by 25% from some 5.24 billion in 2014 to about 3.93 billion PSBs in 2015. [10].

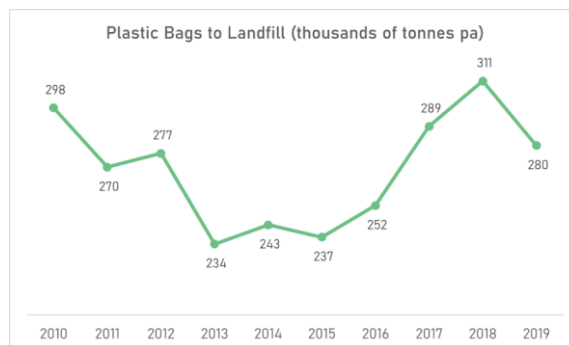


Figure 4: Graph of PSB Disposal from 2010 to 2019 in HK [11]

However, Figure 4 shows that from 2016 to 2018, PSB disposal gradually increased once again, suggesting that the deterrence effect of the second phase of PSB charging may not be as practical in the long term after HK people gradually became accustomed to the cost of PSB charging, in addition to people exploiting the exemption of PSB charging for specific uses.

For the second phase of the PSB Charging Scheme, during a study of the HK population's thoughts regarding the PSB Charging Scheme, Ma et al. [12] found that among 100 residents, 62% of the respondents considered that this policy could encourage them to reduce the generation of plastic waste. In contrast, 30% of respondents thought that this policy could not encourage them to reduce plastic waste production. This suggests that PSB Charging has the effect of discouraging people from using PSB in their daily lives. However, in the same study, 54% of the respondents thought the amount of plastic bag levy should be increased. This study suggests that while the policy influenced decreasing PSB usage, the procedure's effectiveness can be questioned as HK people think that the dissuading effect of the PSB charging is too low due to the relatively low cost of 0.5 HKD to them.

As such, the HKSAR government increasing the cost of PSB to 1 HKD was an effective measure to improve the efficacy of the PSB Charging Scheme. In addition to increasing the price of PSBs, limiting the exemptions for plastic bags can also enhance the policy to prevent people from exploiting exemptions and overusing PSBs.

Due to the third phase of PSB charging being in effect for less than a year, data on public perception of the new updates and the PSB disposal rate cannot be gathered. Thus, empirical information has yet to be published. However, based on the reception and data collected from the full implementation of the PSB charging scheme, it may be suggested that the short-term effect of the amendments to the project will be effective at reducing the PSB disposal rate per year, where the population would gradually become accustomed to the cost of PSB and go back to using them, as many people think that 1 HKD is still too small to deter people from using them, as in a survey by the Kowloon East Office of the Hong Kong Federation of Trade Unions (HKFTU) and World Wide Fund for Nature Hong Kong (WWF) stated that under the 50 cent levy, over 24.8% of respondents would still pay for a plastic bag. [13]

Overall, research and evidence suggest that while the PSB Charging Scheme was effective as a short-term method to reduce the PSB disposal rate, it would gradually decrease in effectiveness over the longer term due to the public being accustomed to the cost and the levy needing to be more significant.

Commented [YS1]: <https://www.wwf.org.hk/en/724803/Existing-Plastic-Bag-Levy-Losing-Effectiveness-Government-should-review-and-promote-environmentally-friendly-practices#:~:text=Under%20the%20current%2050%2Dcent,r espondents%20would%20take%20plastic%20bags.>

MSW Charging Scheme

Overview of MSW Charging Scheme

The MSW Charging Scheme, which will be implemented on 1 April 2024, is a significant part of the plan by the HKSAR government to achieve their medium-term targets of reducing the rate of MSW disposal per capita and increasing the recovery rate of MSW, eventually phasing out the need to dispose of MSW in landfills by 2035.



Figure 5: Logo for MSW Charging Scheme [14]

This policy would target all aspects of MSW: commercial, industrial, and domestic waste. The charge would be levied through two modes: charging by pre-paid designated garbage bags and charging by weight-based "gate fee" at refuse transfer stations or landfills [15], which only applies to waste that does not fit in the bags.

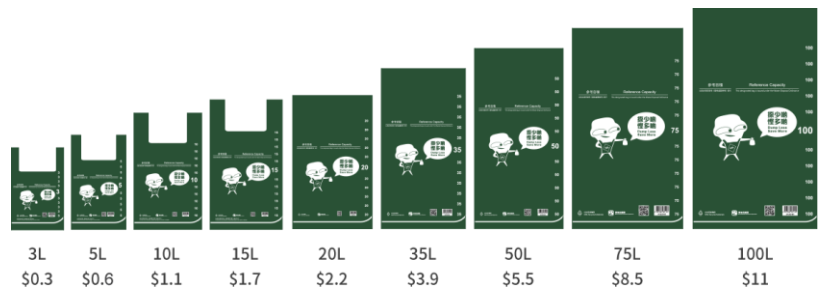


Figure 6: Plastic Bag Sizes for MSW Charging and their respective costs [16]

Effectiveness and Public Reception

During a survey conducted by the Civic Exchange in 2021, the public approval of MSW charging was 68%, and 50% believed it would positively impact reducing HK's waste disposal. [17] This indicates that the MSW charging system is well received by most of the public, which suggests that the population would be inclined to follow the MSW charging scheme.

However, there are some concerns by multiple groups. Street cleaners were concerned about the extra workload as they questioned if people would dump their garbage on the streets [18]. In addition, as HK's housing is mainly based on apartment housing, where there are multiple flats on each floor, it will be challenging to determine who did or did not use the designated trash bags.

Utilising the effectiveness of other cities' MSW Charging Schemes, a prediction of the scheme's effectiveness can be found using other cities that have already implemented a form of MSW Charging.

Taipei's Pay As You Throw (PAYT) system is a policy that charges waste according to the amount of it through the levy on certified plastic bags, similar to HK's plan to acquire the tax through garbage bags, and the bigger the size, the higher the price. The waste charging system in Taipei resulted in a decrease in garbage generated to 1,008 tons from 2970 tons from 2000 to 2011 and a 65% reduction in waste generation per capita [19]. This indicates that an MSW Charging Scheme will reduce the disposal of MSW if implemented correctly.

Although Taipei's PAYT system effectively reduced waste, one significant difference between HK and Taipei is the difference in recycling efforts. In Taipei, recyclables comprise 55% of trash collected from households and commerce and 77% of industrial waste [20]. Recycling in Taipei is much more established and taken seriously. Conversely, recyclables comprise only 31% of HK's MSW, highlighting that HK may not take in all the trash diverted towards recycling after implementing the MSW Charging Scheme.

Recommendations

Considering the potential problems that can occur later for both the PSB Charging Scheme and the MSW Charging Scheme, if such Charging Schemes are implemented, a robust and well-developed supporting infrastructure must be designed to ensure that the scheme can reach its maximum potential and reduce the potential side effects of it.

For recommendations that can benefit both the PSB Charging Scheme and the MSW Charging Scheme, HK should develop more vigorous recycling programs to improve the collection, provide more comprehensive education about waste reduction and recycling, and implement incentives to reduce waste.

Currently, HK needs more collection facilities for recyclables. HK heavily relies on the community's initiative to recycle, as citizens need to go out of their way to collect and transport recyclables to designated spots for recyclables in their residential apartment buildings or at GREEN @ COMMUNITY Stores.



Figure 7: Recycling Bins in Residential/Business Buildings [21]

Commented [YS2]: https://www.housingauthority.gov.hk/en/about-us/publications-and-statistics/housing-dimensions/article/20210416/20210416_newmoves3.html



Figure 8 Interior of a GREEN COMMUNITY Recycling store [22]

However, with the high amount of waste expected to be diverted to recycling stations and bins, there is a concern that these recycling spots can only intake some trash. Thus, increasing the number and storage of recyclable sites is critical to the continued success of the MSW Charging Scheme. Another way to relieve the future pressure on these areas is to implement measures like Taipei's, such as providing recycling trucks with garbage trucks to pick up the recyclable waste [23] to transport to recycling facilities.

The success of waste reduction through the implementation of MSW Charging and PSB Charging can only be achieved through the public's cooperation. Education has proven to be an effective method to raise public approval and acceptance of recycling. Educating people on how and why MSW Charging is implemented helps raise awareness of HK's current waste problem. It increases public participation in these goals, such as developing the habit of citizens recycling and promoting recycling to the people.

HK must take a multi-pronged approach to MSW Charging, as only implementing MSW Charging may cause adverse effects elsewhere. HK only succeeds in the zero-landfill goals through preparation to mitigate these future problems.

Commented [YS3]: <https://www.wastereduction.gov.hk/en-hk/waste-reduction-programme/greencommunity>

Commented [YS4]: <https://doi.org/10.1016/j.wasman.2005.10.005>

Conclusion

In conclusion, HK still has a long way to go to become an eco-city. While HK's waste production ranks foremost in most rankings, the HK government recognises the essential need for HK to reduce its wasteful habits. Exploring the current situation of HK's waste management system, the execution of "polluter-pays" policies has assisted in relieving the pressure of waste on HK's landfills. The accomplishment of being a city with zero landfills is a daunting task that would take more than a decade to achieve. Only one or two cities have reached this goal, and as HK develops even more, this will become increasingly difficult.

With the enhancement of the PSB Charging Scheme and the implementation of MSW Charging, HK could improve its waste management even further. However, many potential concerns and problems may occur without considering different scenarios and factors. Even though HK as a city has the unique circumstance of being a place with high population density and low land availability, numerous solutions can be implemented to increase the effectiveness of these policies by taking references from different cities.

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